

MicroDuck 单臂 v1-B 工程文档

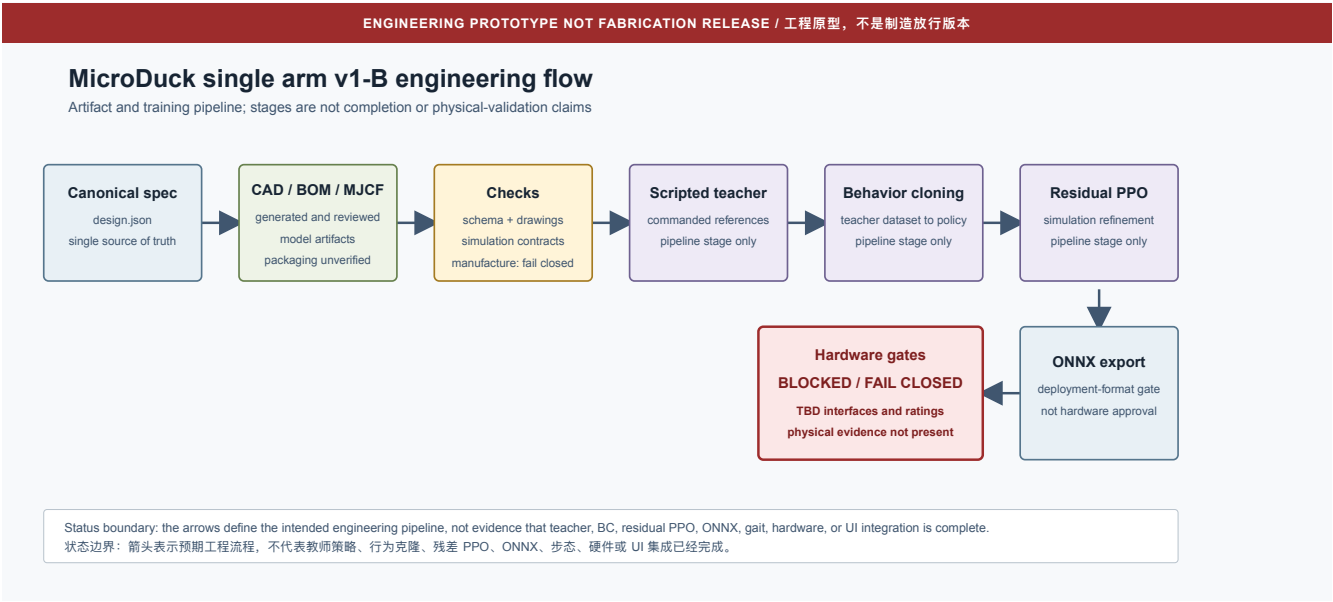


Figure 1: MicroDuck arm v1-B engineering flow

MicroDuck single arm v1-B engineering package

ENGINEERING PROTOTYPE NOT FABRICATION RELEASE 工程原型，不是制造放行版本

This package documents one five-actuator arm cartridge: four pose axes plus one rotary geared gripper. It provides editable OpenSCAD, review drawings, a logical electrical netlist, a bilingual BOM, a commanded test SOP, and a fail-closed acceptance record.

本文件包描述一个五执行器单臂卡匣：四个姿态轴加一个旋转齿轮夹爪。内容包括 可编辑 OpenSCAD、评审图、电气逻辑网表、双语 BOM、指令式测试 SOP 和默认 失败关闭的验收记录。

Design boundary / 设计边界

- Nominal axes: shoulder-elbow 55 mm, elbow-wrist 50 mm, wrist-TCP 30 mm.
- Five XL330-M288-T candidates at 18 g each.
- Motor envelope 20 x 34 x 26 mm is clearance-only. The available visual mesh is about 29 mm deep and is not a manufacturing interface.
- The gripper uses two equal external gears at local $y = +/-12$ mm, ratio -1, with 30 mm rotary jaws about parallel +Z axes. Left joint range is $[-0.32, 0]$ rad; right is $[0, 0.32]$ rad; right = -left.
- Central trunk mount target is 20 g; complete arm cartridge target is 160 g. Both are unverified design targets.
- Servo/horn holes, gear tooth form, bearings, fasteners, tolerances, cable exits, and base attachment are intentionally absent.
- Motor envelopes follow the canonical local body-frame centers. Gripper motor center is $[7, 0, 22]$ mm above the hand frame; wrist motor center is $[25, 0, 20]$ mm in the forearm frame. Overall packaging, interference, cable routing, and table clearance remain unverified.
- 标称轴距：肩肘 55 mm、肘腕 50 mm、腕部到 TCP 30 mm。
- 五颗 XL330-M288-T 候选电机，每颗 18 g。
- 20 x 34 x 26 mm 仅是间隙包络。现有可视网格深度约 29 mm，不是制造接口。
- 夹爪为两只等尺寸外啮合齿轮，本地中心位于 $y = +/-12$ mm，传动比 -1，30 mm 旋转夹指绕平行 +Z 轴运动。左关节范围 $[-0.32, 0]$ rad，右关节范围 $[0, 0.32]$ rad，并满足 right = -left；不使用理想滑块。
- 中央躯干安装件目标 20 g，完整手臂卡匣目标 160 g；两者均未验证。
- 舵机/舵盘孔、齿形、轴承、紧固件、公差、走线出口和根部连接均故意留空。
- 电机包络采用规格中的本体局部坐标。夹爪电机中心在手部坐标系 $[7, 0, 22]$ mm，腕部电机中心在前臂坐标系 $[25, 0, 20]$ mm。总体包装、干涉、走线和桌面间隙均未验证。

Canonical source / 唯一规格源

hardware/microduck-arm-v1/spec/design.json is owned by the design lead. The generator reads it and rejects values inconsistent with v1-B. No duplicate specification is stored in this documentation slice.

hardware/microduck-arm-v1/spec/design.json 由设计负责人维护。生成器读取该文件，并拒绝与 v1-B 不一致的数值。本文件包不保存冲突的规格副本。

```
python3 hardware/microduck-arm-v1/cad/generate_engineering.py
python3 hardware/microduck-arm-v1/cad/generate_engineering.py --manufacture-check
```

The second command is expected to return nonzero until every TBD release gate has objective evidence. A zero exit would require a separately reviewed future revision.

第二条命令在所有 TBD 放行门槛获得客观证据前应返回非零。只有经过独立评审的 未来修订才可允许成功。

Engineering and training flow / 工程与训练流程

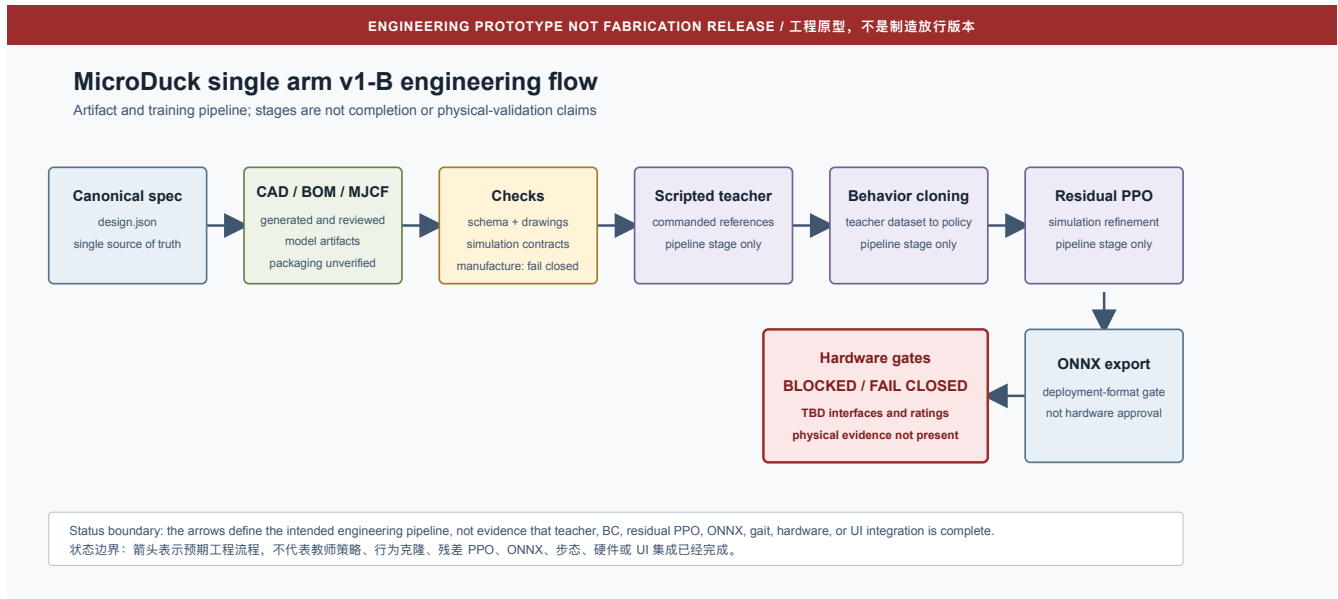


Figure 2: MicroDuck arm v1-B engineering flow

The current engineering CLI regenerates the CAD parameters, mechanical and electrical review drawings, netlist, and fail-closed manufacture record from the canonical specification:

当前工程命令行从唯一规格重新生成 CAD 参数、机械和电气评审图、网表以及失败关闭的制造检查记录：

```
python3 hardware/microduck-arm-v1/cad/generate_engineering.py
python3 hardware/microduck-arm-v1/cad/generate_engineering.py --manufacture-check
cd rlx
uv run --with pytest pytest -q tests/test_microduck_arm_v1_hardware.py
```

The intended downstream sequence is scripted teacher, behavior cloning, residual PPO, ONNX export, and then independent hardware release gates. These are pipeline stages, not claims of completed training or integration. The leader-owned [English results](#) and [Chinese results](#) are the status and evidence records when present. This package does not claim completed gait, hardware operation, or UI integration.

预期后续顺序为脚本教师、行为克隆、残差 PPO、ONNX 导出，再进入独立硬件放行门槛。这些只是流程阶段，不表示训练或集成已经完成。负责人维护的 [英文结果](#) 和 [中文结果](#) 在文件存在时作为状态和证据记录。本文件包不声明步态、硬件操作或 UI 集成已经完成。

Build the bilingual engineering books / 构建双语工程书

```
bash docs/microduck-arm-v1/build-book.sh
```

The script requires installed pandoc, xelatex, and rsvg-convert. It fails clearly until both leader-owned result files exist, then writes:

- docs/microduck-arm-v1/MicroDuck-Arm-v1-B-Engineering.en.pdf
- docs/microduck-arm-v1/MicroDuck-Arm-v1-B-Engineering.zh-CN.pdf

脚本需要已安装的 pandoc、xelatex 和 rsvg-convert。在两份负责人结果文件都存在之前，脚本会明确失败；文件齐备后生成上述英文和中文 PDF。

Modeling modes / 建模模式

Fixed-root bench mode is the only intended first physical integration: the arm root is fixed to a separately verified bench fixture, with a low workspace, catch provision, conservative software limits, and an accessible power disconnect. This package does not design or approve that fixture.

Floating mode is a simulation assumption only. It may use provisional mass placeholders to study system behavior, but it does not prove that a MicroDuck can carry, power, balance, or safely control this arm. No attachment to the original robot body or battery positive is approved.

固定根部台架模式是首个实体集成的唯一预期方式：手臂根部固定到另行验证的台架夹具，采用低位工作区、承接措施、保守的软件限位和可触及的断电装置。本文件包不设计也不批准该夹具。

浮动模式仅是仿真假设。可使用临时质量占位值研究系统行为，但不能证明 MicroDuck 能承载、供电、平衡或安全控制该手臂。禁止把手臂接到原机电池正极，也未批准任何机身安装。

Files / 文件

- hardware/microduck-arm-v1/cad/microduck_arm_v1_b.scad: editable concept.
- hardware/microduck-arm-v1/cad/mechanical-dimensions.svg: dimension review.
- hardware/microduck-arm-v1/cad/mechanical-dimensions.pdf: PDF export.
- hardware/microduck-arm-v1/cad/manufacture-check.json: fail-closed gates.
- hardware/microduck-arm-v1/electrical/wiring.svg: bench wiring concept.
- hardware/microduck-arm-v1/electrical/wiring.pdf: PDF export.
- hardware/microduck-arm-v1/electrical/circuit.svg: logical circuit boundary.
- hardware/microduck-arm-v1/electrical/circuit.pdf: PDF export.
- hardware/microduck-arm-v1/electrical/netlist.json: unresolved-safe netlist.
- BOM.md, SOP.md, HARDWARE-ACCEPTANCE.md: bilingual controlled records.
- engineering-flow.svg: intended pipeline with the hardware gate blocked.
- build-book.sh: English/Chinese PDF book builder; requires leader results.

Official references used for candidate identity and connector semantics:

- ROBOTIS XL330-M288-T e-Manual: <https://emanual.robotis.com/docs/en/dxl/x/xl330-m288/>
- ROBOTIS U2D2 e-Manual: <https://emanual.robotis.com/docs/en/parts/interface/u2d2/>

These references do not release the project-specific mechanical or electrical interfaces.

MicroDuck arm v1-B BOM / 物料清单

ENGINEERING PROTOTYPE NOT FABRICATION RELEASE
No purchasing or substitution approval / 不代表采购或替代批准

Quantities are for one arm. A named candidate is not an approved assembly until all compatibility and release gates pass.

数量按单臂计算。候选型号只有在兼容性和放行门槛全部通过后才可成为批准组件。

ID	Qty	Candidate or requirement / 候选或要求	Status / 状态	Required evidence / 必需证据
M1-M5	5	ROBOTIS XL330-M288-T, 18 g each	Candidate exact variant / 精确型号候选	Purchased label, readable identity/firmware, unique bus ID, connector inspection, measured fit
TRUNK-BLANK	1	Central trunk mount, 20 g target	Design target, unmeasured / 设计目标, 未测	Released interface, weighed article, fixed-root fixture load review
ARM-CARTRIDGE	1	Complete cartridge, 160 g target	Design target, unmeasured / 设计目标, 未测	Complete weighed assembly and center-of-mass record
LINK-BLANKS	1 set	55 mm and 50 mm axis-distance solid blanks	Concept only / 仅概念	Official mating definition, coupon fit, material/process/load review
WRIST-GRIPPER	1	30 mm wrist-TCP; equal gears ratio -1; jaw centers y=+/-12 mm; 30 mm jaws; left [-0.32,0] rad, right [0,0.32] rad, right=-left	Concept only / 仅概念	Gear module/teeth/backlash/material, shaft support, retention, pinch review, grip calibration
U2D2	1	ROBOTIS U2D2, bench TTL candidate	Bench candidate; mobile TBD / 台架候选, 移动方案待定	Correct TTL port, stable enumeration, communication test, mobile architecture review

ID	Qty	Candidate or requirement / 候选或要求	Status / 状态	Required evidence / 必需证据
HUB1	1	Compatible TTL/ power hub	Unselected / 未选型	Exact part, topology, current path, branch protection, connector ratings
BAT1	1	Separate arm battery source for regulated 5 V domain	Unselected / 未选型	Chemistry, cells, capacity, fault current, enclosure, charger and transport controls
BMS1	1	Pack-compatible BMS/protection	Unselected / 未选型	Exact pack compatibility and voltage/current/fault/temperature ratings
F_MAIN	1	Main arm feeder protection	Unselected / 未选型	Part number, rating/curve, conductor/connector coordination, measured transients
F_M1-F_M5	5	One protected VDD branch per actuator	Unselected / 未选型	Part numbers, ratings/curves, branch fault verification
SW_DC	1	Manual DC disconnect or appropriate power pole	Unselected / 未选型	DC interrupt rating at measured source fault conditions; reset behavior
HARNESS	1 set	GND/VDD/DATA harness with strain relief	Unselected / 未选型	Pin-by-pin continuity, wire/insulation/flex ratings, polarity keying, bend/strain review
FIXTURE	1	Fixed-root bench fixture	Outside released CAD / 不在放行 CAD 内	Table/fixture dimensions, fasteners/clamp, overturning/load test, guarded workspace

Provisional simulation mass assumptions / 临时仿真质量假设

The following values may be retained in a model only as **unmeasured placeholders**. They are not BOM facts, procurement specifications, measured mass, payload evidence, or physical validation:

下列数值只能作为模型中的**未测量占位值**保留。它们不是 BOM 事实、采购规格、实测质量、载荷证据或实体验证：

Modeling placeholder / 建模占位项	Provisional mass / 临时质量	Provisional center (m) / 临时中心
Retained electronics / 保留电子件	70 g	[-0.010, 0, 0.018]
Separate arm pack / 独立手臂电池包	80 g	[-0.040, 0, 0.015]
Power hardware / 电源硬件	25 g	[-0.020, 0, 0.034]
Sensor shell / 传感器外壳	8 g	[-0.020, 0, 0.059]

Every model and report using these values must label them `unmeasured_placeholder`. Floating-mode conclusions must be rerun after actual component selection, weighing, center-of-mass measurement, and inertia update.

所有使用这些数值的模型和报告必须标记 `unmeasured_placeholder`。实际选型、称重、重心测量和惯量更新后，必须重新执行浮动模式分析。

Non-substitution and prohibited connections / 禁止替代与连接

- Do not substitute M077, XL320, XL430, XC330, or another voltage/protocol variant without a new design review.
- Never connect ARM_5V to original MicroDuck battery positive or body VDD.
- Never connect arm TTL DATA to the original body actuator DATA bus.
- U2D2 is a communication candidate, not proof of servo power delivery.
- No fuse, connector, BMS, pack, or disconnect rating is inferred from a marketplace title or arithmetic stall-current sum.
- 未经新设计评审，不得以 M077、XL320、XL430、XC330 或其它电压/协议型号替代。
- 禁止将 ARM_5V 接到原 MicroDuck 电池正极或机身 VDD。
- 禁止将手臂 TTL DATA 接到原机身执行器 DATA 总线。
- U2D2 只是通信候选，不证明能够给舵机供电。
- 不得根据网购标题或堵转电流算术和推断保险丝、连接器、BMS、电池包或断电器额定值。

MicroDuck arm v1-B commanded test SOP / 指令式测试规程

ENGINEERING PROTOTYPE NOT FABRICATION RELEASE

Procedure definition only; no test in this document has been run or passed.

仅定义流程；本文档不声明任何测试已执行或通过。

This SOP is for a qualified operator after component selection and an independent safety review. It gives commanded test steps, not instructions for assembling mains wiring. Use only a suitably enclosed, protected low-voltage source. Stop on unexpected motion, heating, odor, noise, current, voltage drop, communication loss, looseness, or damage.

本 SOP 供完成选型并通过独立安全评审后的合格操作人员使用。它规定指令式测试步骤，不提供市电装配说明。只使用合适的封闭式、受保护低压电源。出现意外运动、发热、异味、异响、异常电流、压降、通信中断、松动或损伤时立即停止。

0. Record control / 记录控制

1. Record the canonical spec revision, CAD/netlist hashes, operator, observer, date, fixture ID, source/BMS/fuse/connector part numbers, firmware, and software commit.
2. Mark every unperformed item NOT RUN; never infer a pass.
3. Keep fixed-root and floating-mode evidence separate.
4. 记录规格修订、CAD/网表哈希、操作员、观察员、日期、夹具 ID、电源/BMS/保险丝/连接器型号、固件和软件提交。
5. 未执行项目标记 NOT RUN，不得推断通过。
6. 固定根部和浮动模式证据必须分开。

1. Document and dead-system inspection / 文档与断电检查

1. Command no motion. Isolate the arm source and verify the documented isolation method before touching the harness.
2. Confirm exactly five M288 candidates and unique planned IDs.
3. Inspect solid support blanks and verify no one machined TBD hole patterns from the envelope or visual mesh.
4. Measure and record actual motor, horn, fastener, link, gear, jaw, cable, and fixture interfaces. Compare against released drawings only.
5. Verify pin-by-pin harness continuity and all explicit no-connects.
6. 不发送运动指令。隔离手臂电源，并在接触线束前核实文件化的隔离方法。
7. 确认正好五颗 M288 候选及计划中的唯一 ID。
8. 检查实心支撑毛坯，确认无人依据包络或可视网格加工 TBD 孔位。
9. 测量并记录电机、舵盘、紧固件、连杆、齿轮、夹指、线缆和夹具实际接口。
10. 按针脚检查线束连续性和全部明确断开点。

2. Power-domain command sequence / 电源域指令顺序

1. Keep all servo torque disabled in software.
2. Command the source on with current limiting/protection set by the approved electrical test plan, not by this document.
3. Record source voltage/current and each protected branch at idle.

4. Command the disconnect open; verify arm VDD is removed while noting that USB may keep U2D2 enumerated.
5. Reset only after an explicit software-safe state; reset must not command motion or automatically restore torque.
6. 软件保持全部舵机扭矩关闭。
7. 按已批准的电气测试计划设置限流/保护后上电，不使用本文档猜测额定值。
8. 记录电源和各保护支路的空闲电压、电流。
9. 指令断电器断开，验证手臂 VDD 被切断；注意 USB 可能仍使 U2D2 保持枚举。
10. 仅在明确的软件安全状态下复位；复位不得触发运动或自动恢复扭矩。

3. Communications and direction / 通信与方向

1. With torque disabled, command a ping/read of one isolated motor at a time.
2. Record model identity, firmware, ID, voltage, temperature, and error state.
3. Command a very small, low-speed motion for one unloaded axis at a time in the fixed-root fixture and low workspace.
4. Verify sign, limit, zero reference, cable motion, and disconnect response.
5. For the gripper, command jaw A through a small angular increment and verify jaw B rotates oppositely through the equal-gear coupling. Reject translation or independent same-direction rotation.
6. 扭矩关闭时逐颗隔离执行 ping/读取。
7. 记录型号、固件、ID、电压、温度和错误状态。
8. 在固定根部夹具和低位工作区内，每次只对一个空载轴发送极小、低速运动指令。
9. 验证方向符号、限位、零位、线缆运动和断电响应。
10. 对夹爪，指令夹指 A 小角度转动，验证夹指 B 通过等齿轮反向转动。若出现平移或同向独立旋转则拒收。

4. Incremental fixed-root motion / 固定根部递增运动

1. Command one joint at a time over a reduced range, then coordinated motion at reduced speed/acceleration.
2. At each step record commanded/observed position, tracking error, voltage, current, temperature, branch protection state, fixture motion, and cable clearance.
3. Command the disconnect during a preplanned low-energy motion and record the stop behavior; do not claim a safety category from this test.
4. Repeat only within the approved thermal/rest plan.
5. 先逐关节在缩小范围内运动，再以降低速度/加速度执行协调运动。
6. 每步记录指令/观测位置、跟踪误差、电压、电流、温度、支路保护状态、夹具位移和 线缆间隙。
7. 在预先规划的低能量运动中指令断电并记录停止行为；不得据此声明安全等级。
8. 只在已批准的温升/休息计划内重复。

5. Load and shoulder evidence / 载荷与肩部证据

1. Measure the complete cartridge mass and center of mass before any payload.
2. Determine worst-case shoulder gravity torque from measured mass geometry.
3. Obtain continuous-duty capability evidence for the actual actuator, thermal environment, voltage, motion profile, and support arrangement.
4. Require documented continuous capability of at least 3 x the measured worst-case gravity torque before advancing. Stall torque is not acceptable evidence.
5. Add test mass incrementally above a catch provision; the 20 g payload target remains unverified until all acceptance criteria pass.
6. 加载前测量完整卡匣质量和重心。
7. 根据实测质量几何计算肩部最坏重力矩。
8. 对实际执行器、热环境、电压、运动曲线和支撑结构取得连续工作能力证据。
9. 继续测试前，文件化连续能力必须至少为实测最坏重力矩的 3 倍。堵转力矩不能 作为证据。
10. 在承接措施上方逐步增加测试质量；20 g 载荷目标在全部验收通过前仍未验证。

6. Floating-mode restriction / 浮动模式限制

Do not transfer fixed-root results to floating mode. Floating simulation uses unmeasured mass placeholders and requires updated measured mass, center of mass, inertia, attachment compliance, power architecture, balance/control analysis, and a separate test authorization before any physical trial.

不得把固定根部结果直接用于浮动模式。浮动仿真使用未测量质量占位值；任何实体 试验前必须更新实测质量、重心、惯量、连接柔度、供电架构和平衡/控制分析，并取得 单独测试授权。

MicroDuck arm v1-B hardware acceptance / 硬件验收

ENGINEERING PROTOTYPE NOT FABRICATION RELEASE

Current state / 当前状态: NOT RUN - FAIL CLOSED / 未执行 - 失败关闭

This is an evidence template. Empty evidence means failure to release, not an implicit pass. It contains no claim that hardware exists, was operated, or passed.

本文件是证据模板。证据为空表示不能放行，不代表默认通过。本文不声明硬件已经 存在、运行或通过测试。

Gate / 门槛	Required evidence / 必需证据	Current result / 当前结果
Canonical configuration	Spec revision and generated artifact hashes match	NOT RUN
Exact actuator identity	Five purchased XL330-M288-T units identified; firmware and unique IDs recorded	NOT RUN
Mechanical mating release	Official revision-controlled geometry plus physical measurements, fit coupon, tolerance review	BLOCKED - TBD
Motor depth discrepancy	29 mm visual mesh and 26 mm body envelope reconciled without rescaling/inference	BLOCKED - TBD
Structural load path	Material, fasteners, dual-side support/bearings, retention, fixture load evidence	BLOCKED - TBD
Gripper mechanism	Equal external gears, ratio -1, centers $y=\pm 12$ mm, 30 mm rotary jaws; tooth/backlash/retention verified	BLOCKED - TBD
Packaging and clearance	Canonical motor centers checked against real envelopes, cables, fasteners, table, fixture and full motion	BLOCKED - UNVERIFIED
Mass targets	Trunk mount measured against 20 g target; complete cartridge measured against 160 g target	NOT RUN
Shoulder gravity margin	Measured worst-case gravity torque and actual continuous-duty evidence $\geq 3\times$	BLOCKED - EVIDENCE MISSING
Separate 5 V arm domain	Verified no connection to original MicroDuck positive/body VDD/body DATA/USB VBUS	NOT RUN
Pack and BMS	Exact compatible parts, ratings, enclosure, charger and fault review	BLOCKED - TBD
Protection	Main and five branch protective devices coordinated with measured loads and conductors	BLOCKED - TBD
Harness/connectors	Pinout, continuity, polarity keying, ratings, strain relief, flex and routing evidence	BLOCKED - TBD
Disconnect behavior	Correct DC-rated device; open/reset behavior tested without automatic motion	BLOCKED - TBD

Gate / 门槛	Required evidence / 必需证据	Current result / 当前结果
U2D2/hub bench interface	Stable communication and verified topology; no assumption that U2D2 powers motors	NOT RUN
Mobile interface	Compute, communication, power and mounting architecture	BLOCKED - TBD
Fixed-root fixture	Geometry, fasteners/clamp, overturning/load, workspace and catch review	BLOCKED - OUTSIDE CAD
Software limits	Joint signs, zeros, ranges, speed/acceleration/current/temperature responses recorded	NOT RUN
Incremental motion	Unloaded and reduced-range commanded tests completed per SOP	NOT RUN
Payload target	Incremental test through 20 g with thermal/tracking/grip/drop evidence	NOT RUN
Floating mode	Measured mass/inertia/attachment/power/balance evidence and separate authorization	BLOCKED

Acceptance rule / 验收规则

Release requires every applicable row to contain traceable evidence and a reviewed PASS. NOT RUN, TBD, BLOCKED, missing data, estimates, simulation only, or fixed-root evidence reused for floating mode all fail closed.

放行要求每个适用条目都包含可追溯证据并经评审标记为 PASS。NOT RUN、TBD、BLOCKED、缺失数据、估算、仅仿真，或把固定根部证据复用于浮动模式，均按失败关闭。

Sign-off / 签署

Role / 角色	Name / 姓名	Revision / 修订	Date / 日期	Decision / 决定
Mechanical reviewer / 机械评审				NOT REVIEWED
Electrical reviewer / 电气评审				NOT REVIEWED
Controls reviewer / 控制评审				NOT REVIEWED
Test lead / 测试负责人				NOT REVIEWED

MicroDuck Single-Arm v1-B: 实现与验证报告

结论：工程验证原型已落地，整机放行未通过

本交付是可执行的**工程验证原型**，不是可直接加工装配的成品，也不代表全部任务已经通过。当前评估入口为 artifacts/microduck-arm-v1/evaluation-attempt-3/evaluation.json。第1、2次评估属于旧源码历史记录，不可与本次结果混算。

必须区分四种证据：软件测试通过、仿真任务成功、文件/视频完整性通过、实物验收通过。固定根部台架的到达任务成功，不代表整机能够行走、搬运、平衡，也不代表电气或硬件已通过验证。

最终定向软件测试为**137项通过**：新工程包42项、旧机械臂回归95项；有7条ONNX导出/追踪警告。原始记录为 artifacts/microduck-arm-v1/software-tests.log 和 software-tests.xml。这不代表物理任务套件通过。

已实现的工程包

交付项	仓库路径	真实状态
唯一规格源、尺寸、质量和关节顺序	hardware/microduck-arm-v1/spec/design.json	明确标注假设和未知项
BOM、装配/指令式测试规程	docs/microduck-arm-v1/BOM.md、SOP.md	电机型号候选确定，其余采购接口仍有待定项
可编辑 CAD、机械尺寸图	hardware/microduck-arm-v1/cad/	SCAD、SVG、PDF 评审图；无已放行的孔位或 STEP/STL
原理图、接线图、网表	hardware/microduck-arm-v1/electrical/	逻辑拓扑，不是生产 PCB 或已验证线束
整机 MJCF	hardware/microduck-arm-v1/generated/microduck-arm-v1-free.xml	自由根部、真实物体接触
台架 MJCF	hardware/microduck-arm-v1/generated/microduck-arm-v1-fixture.xml	明确固定根部辅助，不冒充移动验证
URDF	hardware/microduck-arm-v1/generated/microduck-arm-v1.urdf	运动学交换，回差/碰撞近似有记录
力矩、重心、工作区、任务验证	microduck_arm_v1/validation.py	实际运行，失败完整保留
BC、残差 PPO、ONNX、策略评估	microduck_arm_v1/learning.py	仿真训练工具，不是上机控制器
证据审计与视频索引	microduck_arm_v1/evidence.py	哈希、轨迹、视频帧完整性检查
硬件验收规程	docs/microduck-arm-v1/HARDWARE-ACCEPTANCE.md	未执行，缺失实测证据禁止放行

设计一致性与必要调整

- **4个姿态轴+1个夹爪=5颗执行器**。不能写成“5个姿态自由度再加夹爪”。保留10颗腿部执行器，总计15维动作、64维观测；旧的14动作/61观测策略不能直接加载。
- v1-B 统一使用五颗 XL330-M288-T 候选，在新工程包内保持电机型号和接口定义一致。这不证明持续扭矩足够，也不表示 M077 替代方案已验证。
- 肩肘、肘腕、腕端标称轴距为55/50/30 mm，合计135 mm；这是几何长度，不是承重半径或平衡工作区。
- 夹爪采用旋转双指、-1 齿轮耦合。MJCF equality 和 URDF mimic 描述的是拟议传动；实际齿轮、回差、轴承和强度仍需设计。
- 模型电机包络采用宽/深/高20×26×34 mm。旧可视网格深度与其不同，不能根据网格或包络猜测安装孔。

- 原模型被移除的颈部子树质量为0.2799274 kg，并非只有空外壳。因此重新加入电子件、电池、电源器件和传感器壳的**未实测质量占位**，而不是把它们从重量账中删除。
- 新机器人模型质量为0.80631578 kg，不含任务物体。这不是实物称重结果；新增质量、重心和包装尚未实测。
- 手臂电源正极与原机身正极保持分域。电池/BMS/DC-DC/保险丝/连接器、再生能量保护、热测试仍未完成。本轮未给实物发送运动指令，也未修改机载控制软件。

实际仿真结果

教师为 IK + 分阶段状态机。评估种子910000、910001；每回合最多1500步、控制周期0.02 s，成功或失败时提前结束。任务物体为仿真10 g载荷，不代表已经取得10 g额定承重能力。

新案例	自由底座	固定台架	实际失败原因或限制
到达目标	0/2	2/2	自由底座跌倒
取放物体	0/2	0/2	环境碰撞／机器人自碰撞
避障移动物体	0/2	0/2	环境碰撞／机器人自碰撞
原地经过路点搬运	0/2	0/2	环境碰撞／机器人自碰撞
收臂行走	0/2	不允许	尚无可工作步态，环境碰撞
搬物行走	0/2	不允许	环境碰撞，整机控制器未验证
合计	0/12	2/8	整机任务放行失败

两次台架到达成功的末端位置误差分别为4.63 mm、4.09 mm，完整旋转误差分别为0.0457 rad、0.0427 rad，均经过一秒保持窗口。这只是两个固定种子的仿真结果，不是精度承诺，也不是40/40验收。

旧实验室的单臂到达、取放、移动、搬运在新模型中有对应场景，但旧形态的通过记录不能转移给新机器人。arms-handover-v1 和 arms-co-carry-v1 属于第二阶段双臂任务，单臂版本不声称实现。绘画、DAgger、颜色覆盖率/精确率、真实接触力资格验证也未在此工程包实现。

验收防作弊约束

每个物理子步出现的禁止接触都会记录。到达任务检查完整目标旋转，不能仅凭一根工具轴方向判断成功。有效抬升后失去抓握视为掉落，只有合法放置阶段才能释放；释放许可可撤销，搬物行走不具备此许可。行走要求近期、直立状态下、有前向进展的交替单足支撑事件。收臂行走检查实际臂姿；搬运检查当前抬升高度、双侧抓握、没有地面/桌面支撑，以及受约束的相对机身物体位置。

默认回合改为30秒，覆盖教师21秒时的释放阶段和1秒保持。更短回合只能明确作为冒烟测试，不能冒充完整搬运评估。目前行走教师仅保持 home 姿态，尚未实现步态。

工作区和力矩筛选

- 在定义的笛卡尔空间盒内采样10000个目标，其中3331个满足受约束的水平工具 IK 条件，2831个同时通过碰撞筛选。
- 平衡安全点数量未知。上述数字不能当作可用工作区，也不证明任意六维姿态可达。
- 500姿态准静态计算表采用20 g TCP载荷假设；3倍筛选因子下，所需合格持续扭矩的采样最大值为0.33063 N·m。表中包含明确标记的不可行姿态。
- 电机实际合格持续能力未知。模型±0.10 N·m限幅是筛选假设，不是厂家持续工况额定值。加速度、工具接触、发热和电池动态均未验证。

原始数据在 artifacts/microduck-arm-v1/workspace.json 和 artifacts/microduck-arm-v1/mechanics/torque-com.csv。

训练与视频证据

训练冒烟流程使用成功的台架到达示范、BC、带外部 IK 教师的残差 PPO，以及独立种子评估。精确检查点、loss、reward、归一化、源码哈希和 ONNX 一致性记录在 artifacts/microduck-arm-v1/learning-smoke/，应读取最终汇总而不是旧 attempts。短训练不等于收敛，也不证明 PPO 超过教师或整机移动策略已成功。

当前策略输入的 TCP、物体和目标位姿来自仿真真值；硬件状态估计尚未实现。残差 ONNX 必须配合相同教师、归一化和动作映射，不能直接交给 Radxa/robotd 驱动实物。

已记录的训练指标

训练种子101，采集8个成功示范回合、643条状态/动作样本。BC训练20轮；残差PPO训练2048个环境步、32次优化更新。独立评估种子为201至205；此到达任务冒烟显式采用250步上限。

控制器	独立评估成功数	平均回报	平均步数
IK/FSM教师	5/5	-0.033414	88.0
BC	0/5	-10.173485	250.0
教师+残差PPO	5/5	-0.020445	87.6

BC 动作 MSE 从 0.097145 下降到 0.00033247，但独立评估全部超时失败。PPO 首/末记录总 loss 为 0.090142/0.048559，不要求单调下降；23 个已结束训练回合平均回报-0.102520。原始 JSONL 同时记录 value loss、policy-gradient loss、entropy、近似 KL、clip fraction、explained variance。5 个小样本配对评估不足以证明 PPO 在统计上稳定优于教师。

ONNX 在另外的 301 至 305 种子观测上通过数值一致性检查：BC 最大绝对误差 $1.55e-7$ ，原始 PPO 残差 $2.24e-8$ ；残差不是独立动作策略。

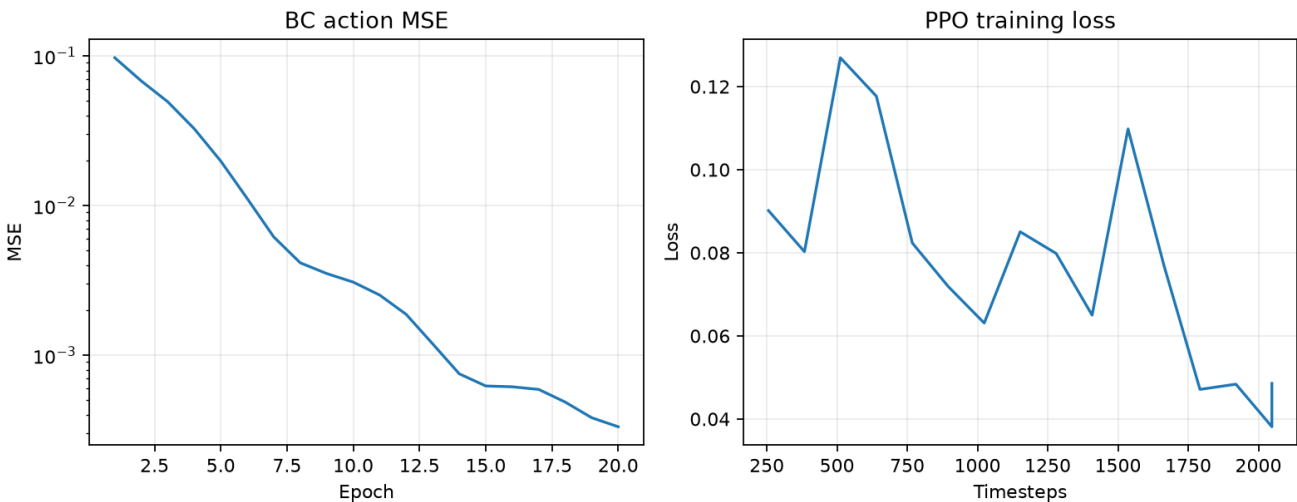


Figure 3: 实际BC和PPO损失曲线

本轮10段 IK/FSM MP4 覆盖全部6个自由底座和4个台架案例，失败视频也保留。打开 [artifacts/microduck-arm-v1/verified-evidence/videos.html](#) 浏览。索引检查每段视频哈希、H.264/25fps格式、包含终止帧的采样帧数及对应轨迹。已查看台架到达与自由底座到达终止画面，分别可见成功和跌倒；视频完整性不等于任务成功。

同一视频库还包含独立种子201的两段实际训练后检查点视频：BC超时、教师+残差PPO成功。**当前共12段视频**，均有轨迹和检查点/文件哈希。训练后视频在 [artifacts/microduck-arm-v1/learned-videos/](#)，没有把教师视频冒充学习策略结果。

关闭执行器和张开夹爪对照保存在 [artifacts/microduck-arm-v1/negative-controls-final/](#)。张开夹爪不是到达任务的有效负对照；如果教师本身无法完成操作，对照失败也不能证明该操作基准有效。

复现方法

在工作区根目录使用已有环境。当前验证版本为 MuJoCo 3.10.0、Stable Baselines3 2.9.0、PyTorch 2.9.1、ONNX Runtime 1.29.0。PDF工具和 ffmpeg/ffprobe 为独立系统工具。本机没有 OpenSCAD CLI，因此未

验证加工网格导出。

```
export PYTHONPATH=.:rlx
PY=rlx/.venv-microduck/bin/python
$PY -m microduck_arm_v1 build
$PY -m microduck_arm_v1 check
$PY -m microduck_arm_v1 torque --out artifacts/my-v1/torque --samples 500
$PY -m microduck_arm_v1 workspace --out artifacts/my-v1/workspace.json --samples 10000
$PY -m microduck_arm_v1 evaluate --out artifacts/my-v1/eval --episodes 2 --max-steps 1500 --videos
$PY -m microduck_arm_v1.evidence controls --out artifacts/my-v1/controls
$PY -m microduck_arm_v1.evidence audit --evaluation artifacts/my-v1/eval/evaluation.json --out artifacts/my-v1,
$PY -m microduck_arm_v1 demonstrate --case reach --mode fixture --episodes 8 --max-steps 250 --out artifacts
```

把 demonstrate 打印的数据集路径交给 train-bc --dataset ... --epochs 20 --out ...。小型残差训练使用 train-ppo --case reach --mode fixture --steps 2048 --max-steps 250 --out ...。使用输出的检查点调用 export --checkpoint ... --out ...。策略评估 API learning.evaluate_policy 要求明确独立种子；最终训练冒烟脚本提供可执行实例。

自由底座任务失败时，evaluate 返回2是预期结果，不能隐去；缺少实测硬件证据时 hardware-check 返回2；制造要求未解决时 generate_engineering.py --manufacture-check 返回3。macOS 离屏渲染可能需要图形服务授权。

下一步工程门槛

1. 实测原机、舵机、安装界面、脚底、电池和电子件，替换质量/惯量占位，解决电机网格尺寸冲突。
2. 修改承力安装件和无碰撞运动路径。当前取物轨迹发生自碰撞，不可通过关闭碰撞制造通过结果。
3. 先实现整机站立平衡，再实现步态并重跑带载自由底座任务。
4. 完成孔位、公差、传动和结构计算，选定并验证电源保护、储能和线束；当前逻辑图不能作为加工授权。
5. 增加传感器估计、时序资格、安全的 robotd/控制中心集成及独立实物验收。旧 viewer 机械臂页面不代表已支持本次64/15契约。
6. 按预登记任务/扰动条件重新训练并比较教师、BC、PPO，再进入受约束台架、站立、接触和低速搬运实测。

本交付终点是可复现原型和明确的失败证据，**不声明全部技能通过、实体能力额定值、加工放行或硬件安全认证。**